



Bedfordview Primary School
Natural Science and Technology Term 2 Controlled Test

Name: _____

Grade 6 : ____

Subject: Natural Science and Technology

Examiner: Mrs N Suliman and Ms Z Kajee

Moderator: Mr J Nash

Date: 2 June 2023

Time Allocation: 1 hour 30 min

INSTRUCTIONS

1. An extra 10 minutes will be provided for reading.
2. Write neatly and legibly.
3. Answer in the spaces provided.
4. This paper consists of 9 pages.

Parent's signature: _____

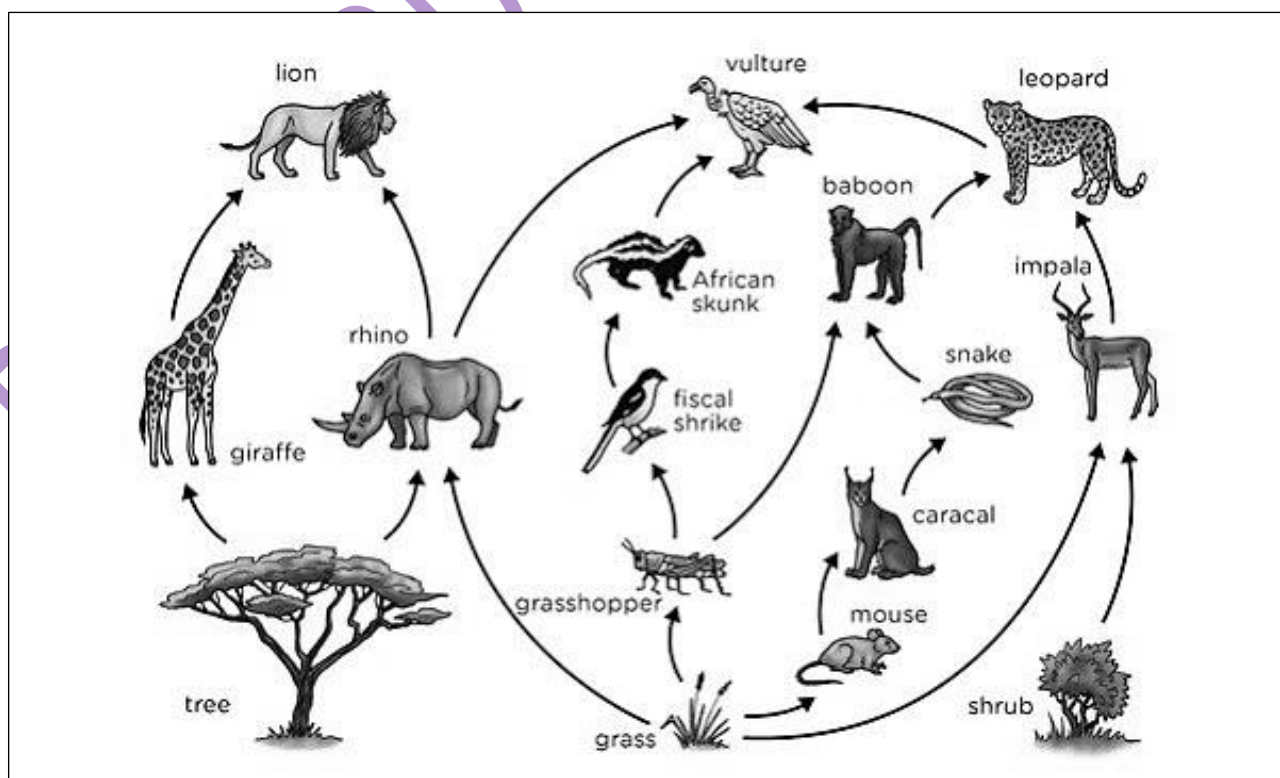
Comments: _____

Question	1	2	Total Section A	3	4	5	6	7	Total Term 2	TOTAL	%
Allocated mark	10	5	15	9	4	9	11	12	45	60	
Learner's mark											
Moderated mark											
1 0 – 29%	2 30 – 39%	3 40 – 49%	4 50 – 59%			5 60 – 69%			6 70 – 79%	7 80 – 100%	

SECTION A: TERM 1 [15 marks]

Question 1: Food Webs (10 marks)

Study the Savannah ecosystem food web and answer the questions:



1.1 Identify three primary consumers in the food web. (3)

Grasshopper, mouse, giraffe, rhino, impala

1.2 What type of carnivore is the vulture? (1)

Scavenger

1.3. Name the ecosystem where the food web is taking place. (1)

Savannah ecosystem

1.4. What do the arrows in the food web show? (1)

The flow of direction (energy)

1.5. Write down a food chain with a leopard in it using the food web above (2)

Grass→mouse→caracal→snake→baboon→leopard→vulture

Grass→impala→leopard→vulture

Grass→grasshopper→baboon→leopard→vulture

1.6. Name the following organisms in the food chain above: (2)

a. Producer – Grass, tree, shrub

b. Secondary consumer - lion, fiscal shrike, snake, baboon

Question 2: Nutrition (5 Marks)

Study the menu below and answer the questions that follow.

LEARNER A	LEARNER B
Breakfast: 1 Apple All bran cereal with milk	Coffee with milk and 3 teaspoons of sugar.
Lunch: A Chicken sandwich on whole wheat bread with lettuce and tomato. Fruit juice Fruit salad	Lunch: Hot chips from the tuckshop A kit kat chocolate 1 can of coke. 1 steak pie.
Supper: Fish Boiled potato Broccoli and cauliflower	Supper: Pizza A can of fanta orange A slice of chocolate cake



2.1. Fill in the missing information in the table below.

Food groups	Foods learner A ate	Foods learner B ate
Food for energy	Apple, whole wheat bread, boiled potato (1)	Sugar, hot chips, kit kat, coke, pie, pizza, cake, steak pie, fanta orange (1)
Foods for growth and repair		Steak

	Chicken, fish	(1)	(1)
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2.2. A balanced diet consists of all 4 food groups as well as water and Fibre .(1)

SECTION B: MATTER AND MATERIALS

QUESTION 3: Multiple choice (9 marks)

Circle or highlight the correct answer.

3.1. Which of the following would best describe air? (1)

- A. The spaces between the particles are very large and the particles are held tightly together.
- B. The spaces between the particles are very small and the particles are held tightly together.
- C. The spaces between the particles are very small and the particles are not held tightly together.
- D. The spaces between the particles are very large and the particles are not held tightly together.**

3.2. The spaces between the particles of a liquid are: (1)

- A. The same as the spaces between the particles of a solid
- B. Smaller than the spaces between the particles of a gas**
- C. The same as the spaces between the particles of a gas
- D. Smaller than the spaces between the particles of a solid



3.3. Fill in the missing word. (2)

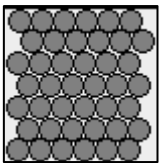
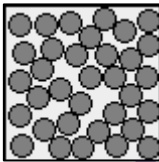
- a) A substance can be poured, and it has very large spaces between its particles. This substance is in the Liquid state.
- b) The particles of a substance only vibrate. This means that the substance is in the Solid state.

3.4. Fill in the missing information in the table below. (5)

DIAGRAM	PHASE	PARTICLE ARRANGEMENT	BEHAVIOUR OF PARTICLES
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(1)	Gas	Particles are far apart with large spaces between particles.	Particles move freely and fast in all directions. Will take up the space of a container and will escape if not sealed (1)
	Solid	Particles are packed tightly together in a regular pattern.	Particles do not move but vibrate (1)
	Liquid (1)	Closely packed but not in a regular pattern. Spaces between particles are very small (1)	The particles are able to move past each other.

Question 4: Mixtures (4 Marks)

Write a paragraph about the hand sorting and decanting separation method Your paragraph must explain the following:

- What type of a mixture is hand sorting? (1)
- How can you separate a mixture using hand sorting? (1)
- What type of a mixture is decanting? (1)
- How can you separate a mixture using decanting? (1)

Remember a paragraph must not have any bullet points.



Do not skip lines in your paragraph.

Complete your planning (key words) below (not for marks) then write your paragraph.

Hand sorting is a solid and solid mixture (2 or more solids) (1) A mixture can be separated through hand sorting by picking the solids out by hand. (1) Decanting is a mixture that consists of two liquids (1) that do not mix well and separate on standing. The oil will float to the top of the water forming a layer above the water, which can then be poured off the water. (1)

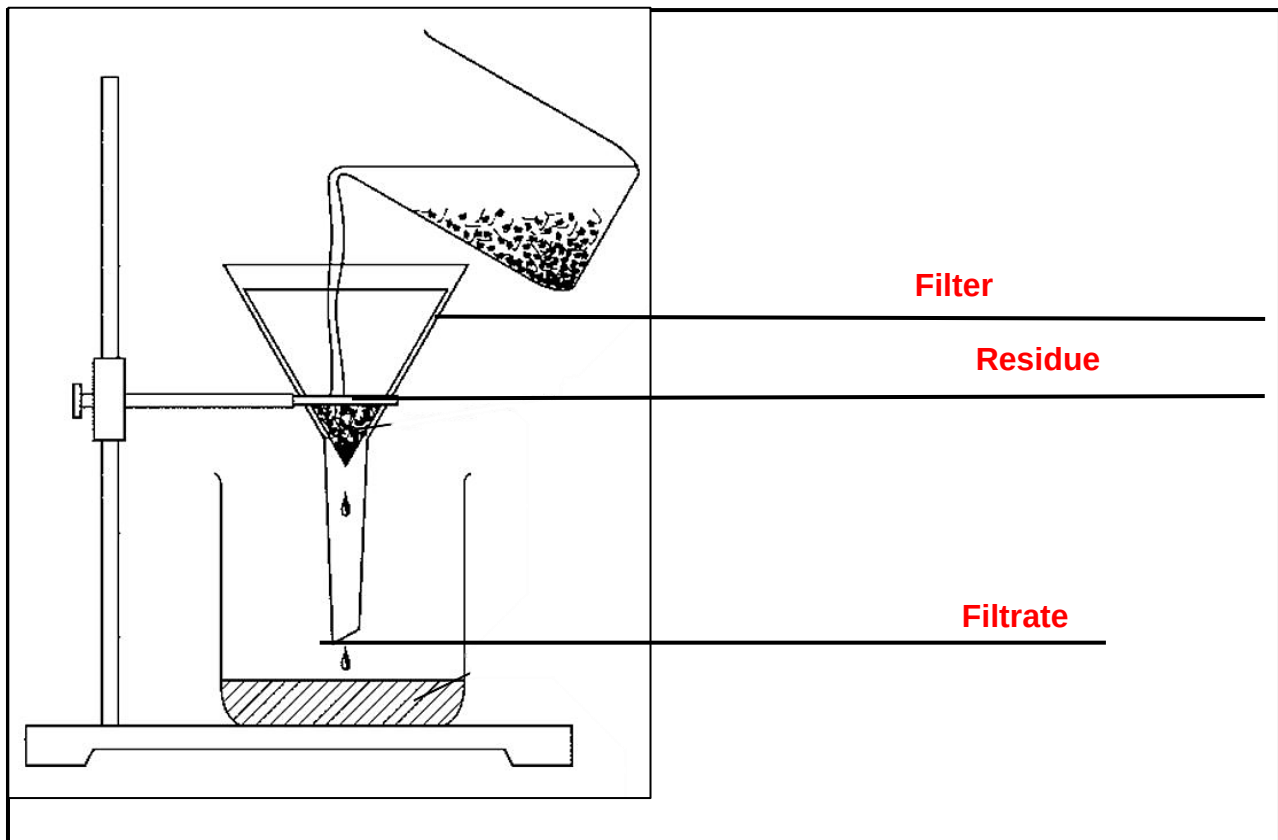


Question 5: Solutions (9 marks)

Answer the following questions.

5.1. Label the diagram.

(3)



5.2. Your mother wants a cup of tea. You need to make sure you have 1 cup of hot water, 2 teaspoons of sugar and 1 tea bag. Write down which one would be:

(2)

a) Solute: Sugar and tea bag

b) Solvent: hot water



Underline/ highlight the correct answer.

c) After stirring the sugar in the tea, you now realise that the sugar has dissolved completely into the hot water. We refer to this as Soluble/ Insoluble. (1)

d) The sweet tea is a saturated/ unsaturated solution. (1)

5.3. Sarah had a bowl of cereal for breakfast. After adding the milk to the cereal with 2 teaspoons of sugar, she then ate her cereal.

a) Write down the solute item in the cereal: Cereal and sugar (1)



b) Sarah had the same cereal the next day, without any sugar. She added milk to the cereal.

Is this mixture, soluble or insoluble?

(1)



Insoluble

Question 6: Dissolving (11 marks)

How temperature affects the rate of dissolving

Fatima and Sam wanted to find out how temperature affects the rate of dissolving.

They used 1 cup of water for all 3 beakers. They used **cold water (beaker 1)**, **warm water (beaker 2)** and **hot water (beaker 3)**. They mixed 1 tsp of salt into each beaker. Look at the table below indicating their results.

Beaker	Temperature of the water °C (Independent Variable)	Time (seconds) for salt to dissolve. (Dependant Variable)
1	10°C	100 seconds
2	30°C	80 seconds
3	50°C	40 seconds

6.1. Name the independent variable in the experiment. (1)

Temperature

6.2. Name the dependant variable in the experiment. (1)

Time

6.3. Name 1 controlled variable (1)

1 tsp of salt

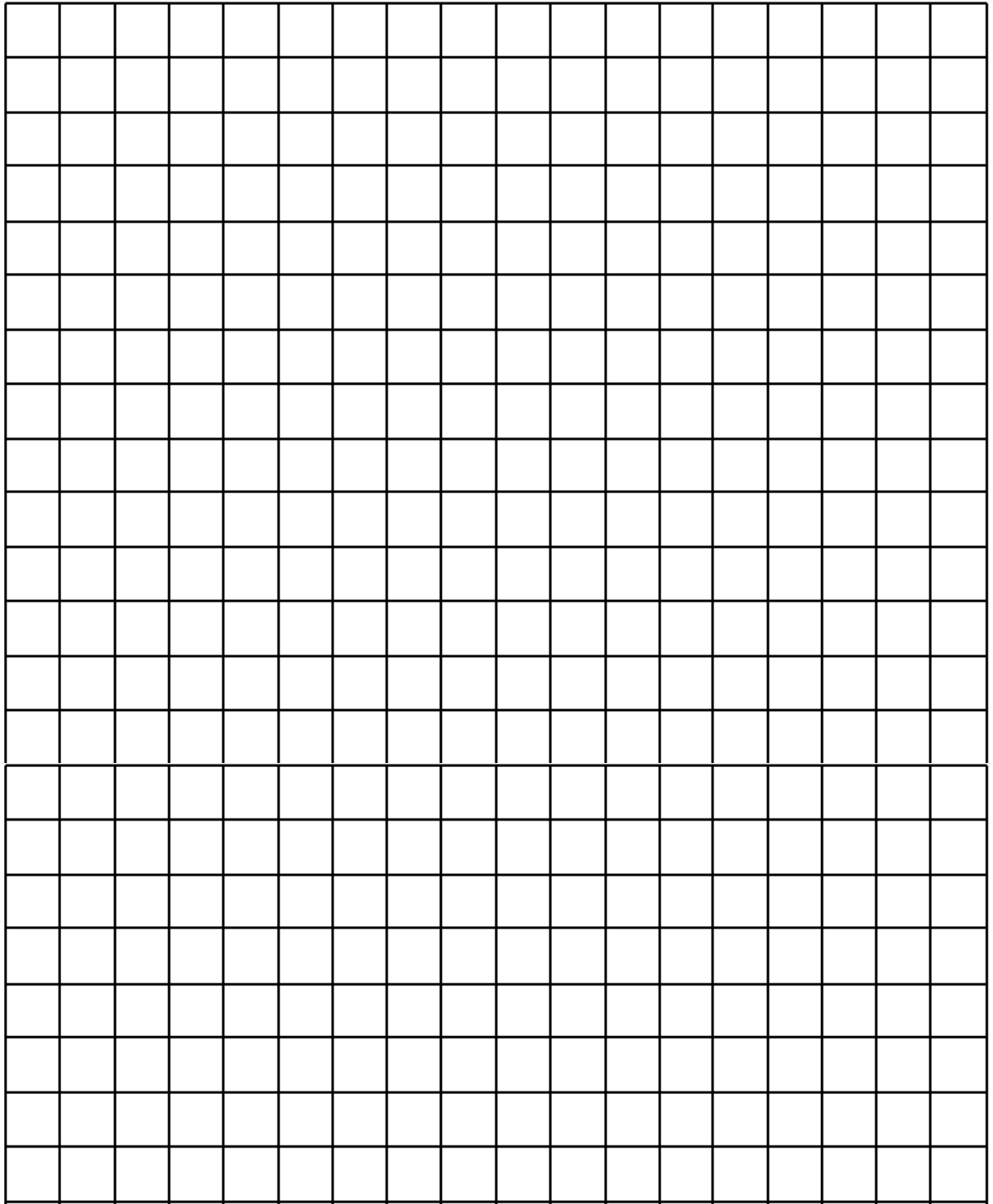
6.4. Draw a line graph on the next page. Remember to write all your numbers in blue pen.

All dots plotted must be in pencil. (8)

Remember to label both the x and the y – axis.

You must give your graph a suitable heading.





Question 7: Water pollution and clean water (12 marks)

Read the newspaper article below and answer the questions that follow.

DURBAN COASTLINE: SEWAGE POLLUTED BEACHES POSE THREAT TO HOLIDAY MAKERS AND THE ENVIRONMENT.

Published: December 11, 2022

Large numbers of businesses in eThekwinini on



South Africa's eastern seaboard – which includes the port city of Durban – rely heavily on tourism. It's a popular holiday destination, only six hours by road from Johannesburg. Millions of people living inland head to the warm coastline over the country's extended December/January school holiday break. But maybe not this year. As Anja du Plessis explains, many of the beaches along the Durban coastline aren't safe to swim in. She explains what's happened, and why.

What's the state of the beaches on the eThekwinini seafront?

The city's beaches are facing an ongoing sewage crisis. Many have been closed along the Durban coastline.

From recorded water quality results, it is quite clear that most of the eThekwinini municipality beaches are currently not safe for public and recreational activities.

Most beaches have critical levels of Escherichia coli. (E. coli). This is a bacteria found in the intestines of healthy people and animals. But high levels of E. coli in water pose a significant threat to human health as well as aquatic and marine environments. The global and South African acceptable standard for E. coli in marine water is 500 counts/100ml. Most beaches along Durban's coastline have critical E. coli levels well above this level and should be considered a public health hazard.

Only three of the nine sampled beaches along the city's extended coastline have acceptable levels of E. coli. They are uShaka, South and Westbrook Main beaches.

About 80% of the pollution in the sea originates on land. Various contaminants (chemicals, nutrients, litter, heavy metals and other toxic substances) are carried by streams and rivers from different land use activities such as farms, industries and urban areas to bays and harbours and finally out to sea.

What's the impact been?

There's a growing risk of major human health hazards and risks, specifically waterborne diseases transmitted via exposure to sewage bacteria and pathogens

8.1. Sewage has been pumped into the ocean in the Natal. What are the problems associated with this? Name 2 problems. (2)

Water-borne diseases, smell, affecting tourism negatively/ affecting the economy

8.2. Name one water-borne disease. (1)

Cholera and diarrhoea

8.3. How could a household clean and purify their water before using it? Name 1 method. (1)

-Boiling the water/ homemade filter

-Chlorine/ household bleach

- Iodine

8.4. How does nature naturally purify water? (1)

Wetlands or by water evaporation

8.5. Name one insoluble pollutant that you can find in the ocean. (1)

Oil spills, plastic, paper, rubber etc.

8.6. Name one method of removing insoluble pollutants from water. (1)

Settling, decanting, sieving and filtering

8.9. Arrange the steps on how municipal water is cleaned before it is used. (5)

Two examples have been done for you.

Steps not in the right order	Steps in the right order (USE LETTERS ONLY)
A. Sedimentation	Step 1: B
B. Storage Dam	Step 2: G
C. Storage and distribution	Step 3: F
D. Disinfection	Step 4: A
E. Filtration	Step 5: E
F. Treatment /Coagulation and flocculation	Step 6: D
G. Screening	Step 7: C

